

Phase 1

Engineering Decisions & Development Log

1. Chunk Size and Overlap

Decision

The Phase 1 ingestion pipeline is designed to divide extracted document text into **150-word chunks with a 30-word overlap** between consecutive chunks.

Technical Rationale

The 150-word chunk size was specified in the Phase 1 ingestion design to divide the extracted aquaculture information into manageable text blocks for subsequent processing and retrieval. The selected size provides a defined unit of text while retaining the information contained within the surrounding context.

A **30-word overlap** is included between consecutive chunks to maintain **contextual continuity**. This is particularly relevant for aquaculture information where an operational value, threshold, condition, or qualification may be described across nearby portions of text. The overlap reduces the likelihood that such information becomes separated completely between adjacent chunks.

The Phase 1 plan specifically identifies the purpose of the overlap as ensuring **contextual continuity** and states that the chunking configuration should use 150-word blocks with a 30-word overlap.

Phase 1 Specification

- **Chunk size:** 150 words
- **Overlap:** 30 words
- **Processing stage:** Document ingestion
- **Purpose:** Maintain contextual continuity during text segmentation

2. Embedding Model

Decision

The Phase 1 pipeline uses **BAAI/bge-m3** as the embedding model for converting the processed text chunks into dense vector representations.

Technical Rationale

After the document text has been extracted and divided into chunks, the chunks need to be represented in a form that can be indexed for retrieval. The Phase 1 plan specifies the use of **BGE-M3** for this embedding stage.

The model was selected in the project specification for its suitability for **cross-lingual Indic/English representation** and **multi-granularity retrieval**. This is relevant to AquaAssist because the knowledge corpus is based on the collected English aquaculture documents, while the broader project is intended to support vernacular interaction.

The embedding stage therefore acts as the representation layer between the processed text corpus and the vector database. The Phase 1 workflow specifies that the text blocks are transformed into dense vectors using BGE-M3 before being indexed into ChromaDB.

Phase 1 Specification

- **Embedding model:** BAAI/bge-m3
- **Input:** Processed text chunks
- **Output:** Dense vector representations
- **Next stage:** ChromaDB indexing
- **Key design consideration:** Cross-lingual Indic/English representation and multi-granularity retrieval

3. Vector Store

Decision

The Phase 1 pipeline uses **ChromaDB** as the local vector store for the generated embeddings.

Technical Rationale

The embeddings generated from the processed text chunks need to be stored in a persistent vector database so that they can subsequently be used during retrieval.

The Phase 1 implementation specifies a **local ChromaDB instance** for this purpose. The project plan identifies ChromaDB as appropriate for **zero-server, offline persistence within local edge-hardware constraints**.

This choice also keeps the vector database as a locally maintained component of the Phase 1 pipeline rather than introducing a separate database server into the architecture. The embedding script is specified to generate the BGE-M3 vectors and index those vectors into the local ChromaDB instance.

Phase 1 Specification

- **Vector store:** ChromaDB
- **Deployment:** Local
- **Purpose:** Persistent storage of generated vectors

- **Architectural requirement:** Offline / local operation
- **Database location:** `data/chroma_db/`

The repository specification further identifies `data/chroma_db/` as a **strictly local** directory rather than a Git-tracked project artifact.

4. Git Policy

Decision

Raw PDF files and the generated **ChromaDB database** are excluded from Git version control. The processed JSON data generated by the ingestion pipeline is permitted to be tracked.

Technical Rationale

The Phase 1 repository structure separates the original documents, processed data, vector database, source code, and documentation into different directories.

The original PDF corpus is maintained outside the Git repository. The Phase 1 plan explicitly states that raw PDF files **must not be committed** and should remain in the shared Google Drive or on individual local machines.

Similarly, the `chroma_db` directory contains generated database files and indexes and is therefore also excluded from version control. The plan specifies that this database should be generated locally by running the embedding pipeline, or transferred separately when manual sharing is required.

In contrast, the processed text output located in `data/processed/` is intended to be tracked. The plan specifically identifies the processed `.json` files as allowed repository data so that team members can work from the same processed text baseline.

Phase 1 Repository Policy

Component	Location	Git Status
Raw PDF corpus	<code>data/raw_pdfs/</code>	Not tracked
Processed text	<code>data/processed/</code>	Tracked
ChromaDB	<code>data/chroma_db/</code>	Not tracked
Python scripts	<code>src/</code>	Tracked
Documentation	<code>docs/</code>	Tracked

This separation is part of the repository architecture defined for Phase 1.